Year 2 Assessed Problems

Semester 1

Assessed Problems 4

SOLUTIONS TO BE SUBMITTED

Wednesday 5th November at 17:00

2 Assessed – 2 Heat engines

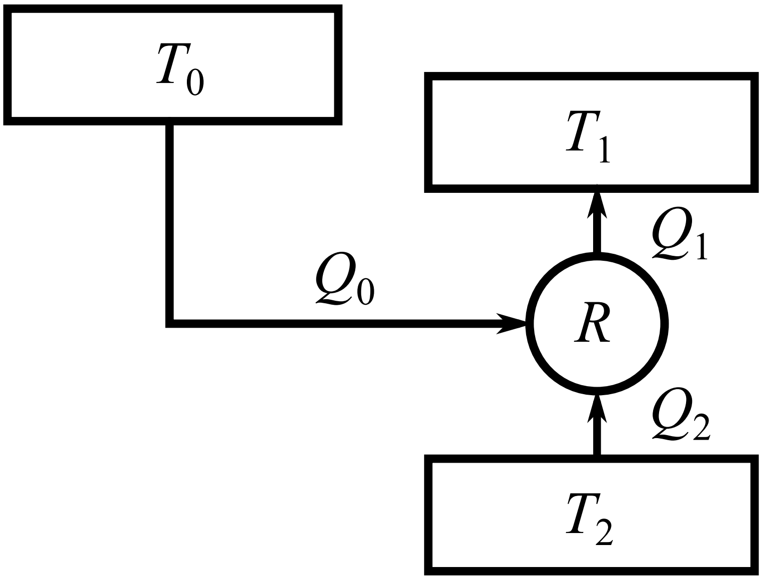

Figure 2.1: The composite refrigerator.

Problem 2.1 **A composite refrigerator**
The diagram, Fig. (2.1), illustrates the principle of an absorption refrigerator R, in which the removal of heat Q_2 per cycle from a cold reservoir at temperature T_2 is driven by absorbing heat Q_0 from a hot reservoir at temperature T_0. Heat Q_1 is released at ambient temperature T_1, as shown, where $T_0 > T_1 > T_2$.

1. Assuming that the refrigerator efficiency is $\eta = Q_2/Q_0 = 3/2$, find an expression for the increase in entropy of the universe as a function of: the pumped heat Q_2 and the temperatures of the reservoirs. **[5]**
2. An actual implementation of such a contraption is a dithermal engine operating between the temperatures T_0 and T_1 and driving a pump working as a refrigerator between the temperatures T_1 and T_2. The efficiencies of the engine and the refrigerator are η_1 and η_2, respectively.
   Find an expression for the efficiency η of the whole device as a function of η_1 and η_2. *Using this expression*, find an expression for η as a function of the temperatures of the reservoirs, assuming that the operation of all the components is reversible.
   Hint: do not confuse the efficiency of a refrigerator with that of a heat pump! **[5]**

2 Generalised functions

★*Problem 2.1* **Use of step, delta functions**
The delta and step functions $\delta(x)$ and $\Theta(x)$ have their usual definitions.

(i) Sketch the function $f(x) = [\Theta(x-2) - \Theta(x-5)]$, indicating carefully on your sketch the values of all relevant coordinates [1]

(ii) Evaluate the integral $\int_{-\infty}^{+7} \mathrm{d}x\, \delta(x-3)\, [\Theta(x-2) - \Theta(x-5)]$. [1]

(iii) Evaluate the integral $\int_{-\infty}^{+5/2} \mathrm{d}x\ \delta(x-3)\, [\Theta(x-2) - \Theta(x-5)]$ [2]

(iv) $\int_{-\infty}^{+1} \mathrm{d}x\ \delta(x-3)\, [\Theta(x-2) - \Theta(x-5)]$. [2]

(v) Evaluate $\int_{-\infty}^{+7} \mathrm{d}x\, x^2\, \delta(x-3)\, [\Theta(x-2) - \Theta(x-5)]$. [2]

(vi) Evaluate $\int_{-\infty}^{+\infty} \Theta(x-1)\, x\, \mathrm{e}^{-x^2}\, \mathrm{d}x$. [2]